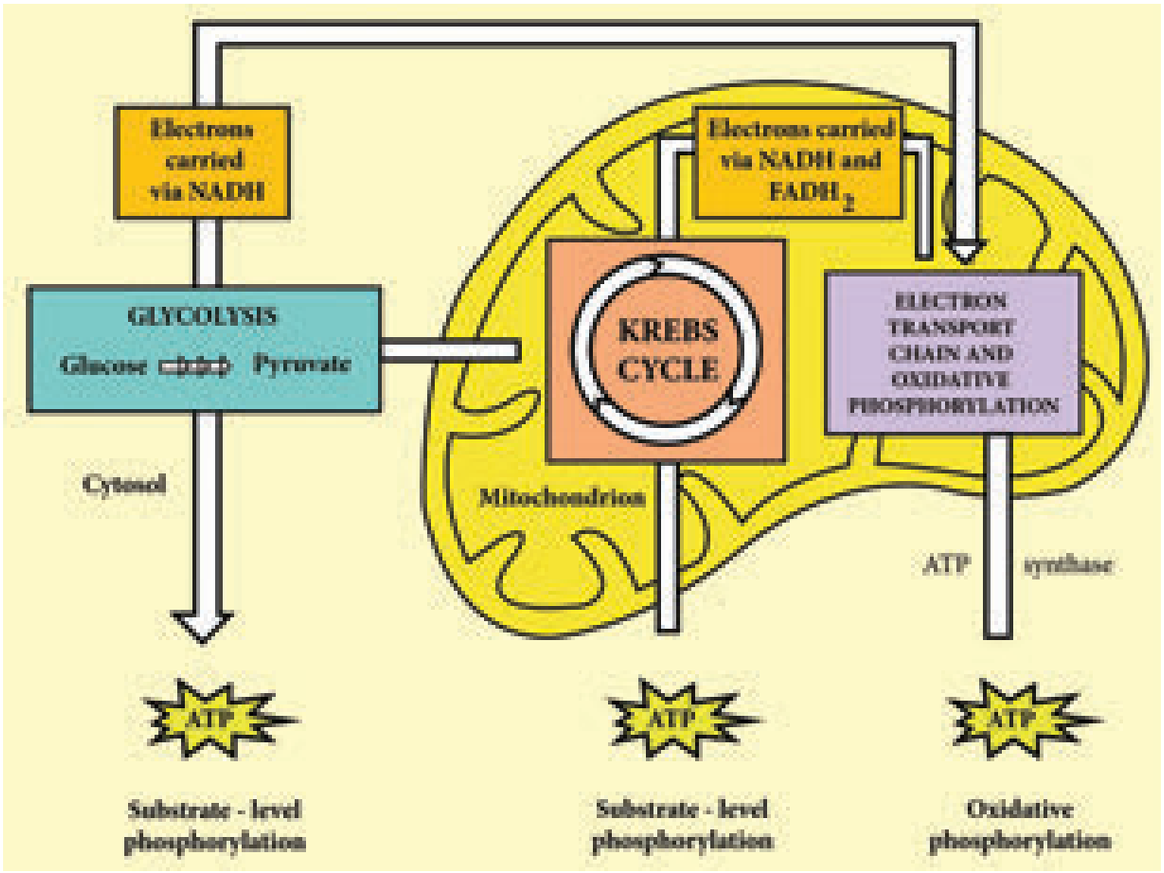


Figure 4.3: Phosphorylation

4.3.3 Photo Phosphorylation

It occurs only in photosynthetic cells which contain light trapping pigments. Example: In photosynthesis, photosynthetic pigment, Chlorophyll is involved in the synthesis of organic molecules especially sugars, with the energy of light from the energy poor building blocks like Carbon dioxide and water. In phototropic bacteria (purple, green sulphur bacteria, Cyanobacteria), photosynthetic pigments bateriochlorophylls are involved in ATP production.

4.4 Carbohydrate Catabolism

Most microorganisms oxidize carbohydrates as their primary source of cellular energy. Carbohydrate catabolism is the breakdown of carbohydrate molecule to produce energy and is therefore of great importance in cell metabolism. Glucose is the most common carbohydrate energy source used by cells.

To produce energy from glucose, microorganism use two general processes namely Respiration and Fermentation.

4.4.1 Cellular Respiration

Respiration is defined as an ATP generating process in which organic molecules are oxidized and the final electron acceptor is an inorganic compound. In aerobic respiration, the final electron acceptor is Oxygen and in anaerobic respiration the final electron acceptor is an inorganic molecule like NO3, SO_4^{2-} other than Oxygen.

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